

Лабораторная работа №9

Администрирование локальных сетей

Дикач А.О.

31.03.2025 г.

Российский университет дружбы народов, Москва, Россия

Информация

- Дикач Анна Олеговна
- НПИбд-01-22 (1132222009)
- Российский университет дружбы народов
- <https://github.com/chelibos?tab=repositories>

Изучение возможностей протокола STP и его модификаций по обеспечению отказоустойчивости сети, агрегированию интерфейсов и перераспределению нагрузки между ними.

Выполнение лабораторной работы

Сформировала резервное соединение между коммутаторами 1 и 3. сделала порт g0/2 транковым, соединила коммутаторы и сделала порты транковыми

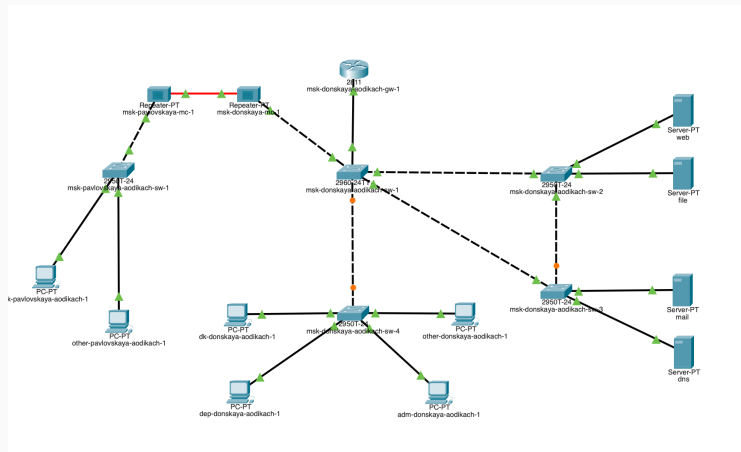


Рис. 1: Изменения в топологии

```
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-sw-3(config)#int g0/2
msk-donskaya-aodikach-sw-3(config-if)#switchport mode trunk
msk-donskaya-aodikach-sw-3(config-if)#exit
msk-donskaya-aodikach-sw-3(config)#exit
msk-donskaya-aodikach-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-aodikach-sw-3#wr m
Building configuration...
[OK]
```

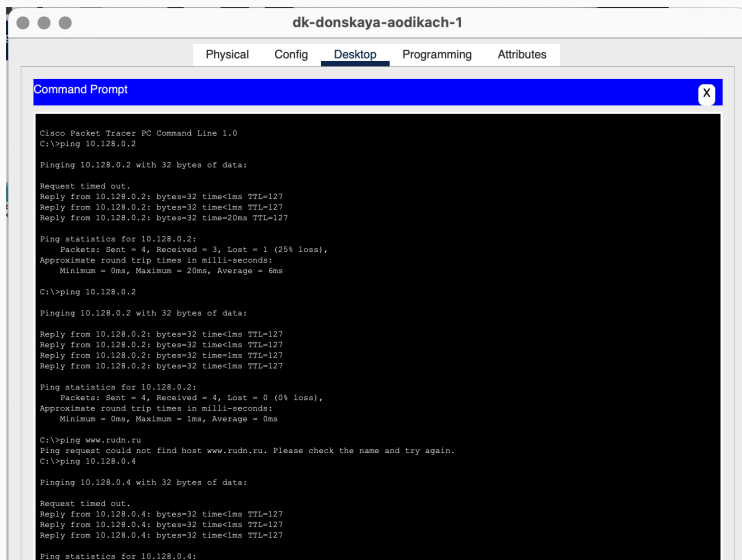
Рис. 2: Настройка портов

```
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-sw-4(config)#int f0/23
msk-donskaya-aodikach-sw-4(config-if)#switchport
UCDP-6-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/23 (104), with msk-donskaya-aodikach-sw-1
FastEthernet0/23 (1).

% Incomplete command.
msk-donskaya-aodikach-sw-4(config-if)#switchport mode trunk
```

Рис. 3: Настройка портов

Пропинговала серверы mail и web. Движение пакетов происходит через коммутатор msk-donskaya-aodikach-sw-2



```
dk-donskaya-aodikach-1
Physical Config Desktop Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time=20ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 6ms

C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127
Reply from 10.128.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping www.rudn.ru
Ping request could not find host www.rudn.ru. Please check the name and try again.
C:\>ping 10.128.0.4

Pinging 10.128.0.4 with 32 bytes of data:

Request timed out.
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.4:
```


На коммутаторе msk-donskaya-sw-2 смотрю состояние протокола STP для vlan 3

```
msk-donskaya-aodikach-sw-2#show spanning-tree vlan 3
VLAN0003
  Spanning tree enabled protocol ieee
  Root ID    Priority    32771
             Address     0001.6328.64D1
             Cost        23
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32771  (priority 32768 sys-id-ext 3)
             Address     0001.9695.BC58
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface    Role  Sts Cost      Prio.Nbr Type
-----
Gi0/1        Root  FWD  4          128.25  P2p
Gi0/2        Desg  FWD  4          128.26  P2p
Fa0/1        Desg  FWD  19         128.1   P2p
Fa0/2        Desg  FWD  19         128.2   P2p

msk-donskaya-aodikach-sw-2#
```

Рис. 5: Состояние протокола

В качестве корневого коммутатора STP настраиваю коммутатор msk-donskaya-sw-1. Убеждаюсь в том что пакеты ICMP проходят от хоста dk-donskaya-1 до mail через коммутаторы msk-donskaya-sw-1 и msk-donskaya-sw-3, а от хоста dk-donskaya-1 до web через коммутаторы msk-donskaya-sw-1 и msk-donskaya-sw-2.

```
Password:
msk-donskaya-aodikach-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-aodikach-sw-1(config)#spanning-tree vlan 3 root primary
msk-donskaya-aodikach-sw-1(config)#
```

Рис. 6: Настройка коммутатора

Настраиваю режим Portfast на интерфейсах коммутаторов

```
Enter configuration commands, one per line. End with CNTL/Z.
mik-donskaya-aodikach-sw-2(config)#
mik-donskaya-aodikach-sw-2(config)#interface f0/1
mik-donskaya-aodikach-sw-2(config-if)#spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
mik-donskaya-aodikach-sw-2(config-if)#exit
mik-donskaya-aodikach-sw-2(config)#interface f0/2
mik-donskaya-aodikach-sw-2(config-if)#spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
mik-donskaya-aodikach-sw-2(config-if)#exit
mik-donskaya-aodikach-sw-2(config)#
```

Рис. 7: Настройка коммутаторов

```
mik-donskaya-aodikach-sw-3enable
Password:
mik-donskaya-aodikach-sw-3#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
mik-donskaya-aodikach-sw-3(config)#interface f0/1
mik-donskaya-aodikach-sw-3(config-if)#spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
mik-donskaya-aodikach-sw-3(config-if)#exit
mik-donskaya-aodikach-sw-3(config)#interface f0/2
mik-donskaya-aodikach-sw-3(config-if)#spanning-tree portfast
Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
mik-donskaya-aodikach-sw-3(config-if)#
```

Рис. 8: Настройка коммутаторов

С помощью команды `ping -n 1000 mail.donskaya.rudn.ru` изучаю отказоустойчивость протокола STP и время восстановления соединения при переключении на резервное соединение

```
C:\>ping -n 1000 mail.donskaya.rudn.ru

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
```

Рис. 9: Работа в терминале

```
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time=1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
```

Рис. 10: Работа в терминале

```
msk-donskaya-aodikach-sw-3(config)#interface f0/1
msk-donskaya-aodikach-sw-3(config-if)#shutdown

msk-donskaya-aodikach-sw-3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

msk-donskaya-aodikach-sw-3(config-if)#exit
msk-donskaya-aodikach-sw-3(config)#interface f0/1
msk-donskaya-aodikach-sw-3(config-if)#no shutdown

msk-donskaya-aodikach-sw-3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
```

Переключаю коммутаторы на режим работы по протоколу Rapid PVST+ и изучаю отказоустойчивость

```
nsk-donskaya-aodikach-sw-3(config-if)#exit  
nsk-donskaya-aodikach-sw-3(config)#spanning-tree mode rapid-pvst
```

Рис. 12: Переключение коммутаторов

```
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127  
Reply from 19.128.0.4: bytes=32 time=1ms TTL=127
```

Рис. 13: Изучение отказоустойчивости

Формирую агрегированное соединение интерфейсов Fa0/20 – Fa0/23 между коммутаторами msk-donskaya-sw-1 и msk-donskaya-sw-4 и настраиваю агрегирование каналов

```
msk-donskaya-aadikach-sw-1(config)#interface range Fa0/20 - 23
msk-donskaya-aadikach-sw-1(config-if-range)#channel-group 1 mode on
msk-donskaya-aadikach-sw-1(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINE-5-CHANGED: Interface Port-channel1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

%EC-5-CANNOT_RESOLVE: Fa0/23 is not compatible with Fa0/20 and will be suspended (dtp mode of Fa0/23 is on, Fa0/20 is off)

%EC-5-CANNOT_RESOLVE: Fa0/23 is not compatible with Fa0/21 and will be suspended (dtp mode of Fa0/23 is on, Fa0/21 is off)

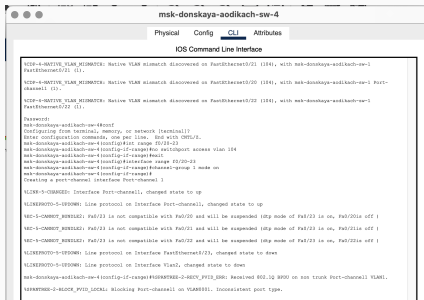
%EC-5-CANNOT_RESOLVE: Fa0/23 is not compatible with Fa0/22 and will be suspended (dtp mode of Fa0/23 is on, Fa0/22 is off)

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down

msk-donskaya-aadikach-sw-1(config-if-range)#exit
msk-donskaya-aadikach-sw-1(config)#interface Port-channel 1
msk-donskaya-aadikach-sw-1(config-if)#switchport mode trunk

msk-donskaya-aadikach-sw-1(config-if)#
```

Рис. 14: Настройка агрегирования каналов



Изучила возможности протокола STP и его модификаций по обеспечению отказоустойчивости сети, агрегированию интерфейсов и перераспределению нагрузки между ними.